

Now, we need help to setup/configure ai capabilities so that Claude can respond with the opinions of an energy expert and provide meaningful insights to Pacific Island Utility CEOs, Managers, Staff, Government Stakeholders, Consultants, Donors, Education researchers etc. What do we need to do to get there and can you guide us step by step?

Before the steps, the one reframe that determines whether this succeeds or embarrasses you:

"Respond with the opinions of an energy expert" is the easy 10%. The dangerous 90% is grounding. A system prompt alone gives you a confident voice that produces *plausible-sounding generic energy commentary* — and the moment it quotes a fabricated diesel-dependency figure to a utility CEO, or invents a "best practice" tariff structure to a donor, your credibility with that audience is gone. For this readership — CEOs, regulators, donors, researchers who will *check* — the architecture's whole job is to make fabrication structurally hard. Expertise has to come from sources you control, with citations, not from the model's memory.

So the real question isn't "how do we give Claude an expert persona," it's **"where does the expertise come from, and how do we prove it."** Everything below is organized around that.

Here's the phased roadmap, with the decision points called out.

Phase 0 — Scope the expertise and its sources (decide before building)

Two decisions gate everything else:

0a. What domains is the "expert" actually expert in? Pick the lanes you'll stand behind. For Pacific utilities the natural set: generation mix & diesel dependency, renewables integration & intermittency, tariff/PPA structures, financial sustainability (cost-reflective tariffs, subsidies), reliability (SAIDI/SAIFI), donor-funded project economics, and regulatory/NDC targets. Scoping this lets you *refuse* gracefully outside it instead of bluffing.

0b. Where does grounded knowledge live? Three tiers, and you want all three eventually:

Source	What it gives you	Effort
Your own data (PRISM benchmarking)	The killer feature — real DHI / PPA-benchmarking KPIs per utility (KPUB, TEC, NUC...). This is what makes it "knows <i>your</i> numbers," not "well-read."	You already have it; needs a query tool
Curated Pacific energy corpus	Pacific Power Association benchmarking reports, IRENA Pacific, SPC/SPREP, FESRIP, World Bank/ADB project docs, country energy roadmaps & NDCs, tariff schedules, PPA templates	Sourcing + rights + ingestion
Claude's parametric knowledge	General energy/utility reasoning to glue the above together	Free, but the part that must never be the source of a <i>number</i>

Decision: confirm you have the rights to ingest the third-party corpus (PPA benchmarking reports especially). If not, you lean harder on your own PRISM data + publicly licensed sources.

Phase 1 — Persona + epistemic guardrails (1–2 days, cheap, do first)

Write the system prompt. Two halves:

- **Voice & audience:** an energy-sector advisor who adapts register by reader (a CEO gets a decision and a tradeoff; staff get the how; a donor gets impact and risk; a researcher gets method and caveats). Make audience an explicit input.
- **Epistemic rules — the important half:** numbers come *only* from retrieved documents or tool results, never from memory; cite every quantitative claim; distinguish "what the data shows" from "my interpretation"; say "I don't have data on that" instead of guessing; surface uncertainty.

This phase alone gets you a demo. It does **not** get you something you'd put in front of a regulator — that needs Phases 2–3.

Phase 2 — Ground it in the corpus (RAG + Citations)

Build the retrieval layer over your curated corpus so answers are anchored in real documents.

- **Ingest & chunk** the Pacific energy corpus; embed and store for retrieval.
- **Retrieve** the top relevant passages per question and pass them to Claude as documents.
- **Use the API's Citations feature** (`citations: {enabled: true}` on document blocks) — Claude then cites the exact spans it used. For this audience, clickable "here's the source" is the difference between an opinion and a defensible insight.
- **Prompt-cache** the stable persona + frequently-used reference docs so you're not paying to re-send them every call.

Trade-off: managed RAG (a vector DB + your pipeline) vs. lighter approaches (curated context windows for a small corpus, since modern context is 1M tokens). If your authoritative corpus is small and stable, you may not need a vector DB at first — you can pass curated docs directly and cache them. Start simple; add retrieval when the corpus outgrows the context.

Phase 3 — Connect to PRISM data (tools) — this is the moat

Wire the same tool-use pattern from our Power BI work: define a tool that queries PRISM's benchmarking KPIs, so the expert can say *"TEC's diesel generation cost per kWh is X, vs the PPA peer median of Y"* with real figures. This is what no generic "energy GPT" can do — it's grounded in your customers' actual operational data. UtilityOS feeding PRISM is the upstream that keeps these numbers fresh.

Same security boundary as before: tool runs server-side, applies the requesting stakeholder's access scope (a consultant for Utility A must not pull Utility B's confidential KPIs — same multi-tenant RLS discipline as the Power BI case).

Phase 4 — Audience modes & output shaping

Make the stakeholder type a first-class parameter (CEO / manager / staff / government / consultant / donor / researcher). Each maps to a register, default length, and what to lead with. Optionally structured outputs for things like a standard "briefing note" format donors expect.

Phase 5 — Evaluation with a real expert in the loop (don't skip)

This is what separates a product from a toy in an advisory domain:

- Build an **eval set**: 30–50 representative questions per stakeholder type, with expert-reviewed "good answer" criteria.
 - Have an **actual Pacific energy specialist** grade outputs for accuracy, not just fluency — fluent-but-wrong is the failure mode that kills trust here.
 - Test the **refusal/uncertainty behavior** explicitly: feed it questions outside scope and where data is missing, and confirm it declines rather than invents.
 - Iterate the prompt + retrieval against the evals.
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Suggested sequencing

Phase 1 (persona) → Phase 3 (PRISM tool — your differentiator) → Phase 2 (corpus + citations) → Phase 4 → Phase 5 running throughout. I'd put the PRISM data tool *before* the big corpus build, because "it quotes our real benchmarking numbers" is the most compelling demo and it reuses the tooling you just built.

Two scoping questions so I can take you into build rather than keep theorizing:

Where should this expert AI live?

A feature inside PRISM

Which do you want to build first?

1 and 3

Good combination — persona for the voice, PRISM tool for the real numbers. Together that's a grounded demo rather than a plausible-sounding one.

PRISM is your Next.js/Drizzle app, so before I write tool code against invented table names, let me map the actual schema and tenancy model so the data tool queries real columns and respects your existing access scoping.

No problem — relaunching the exploration.

This changes my answer materially — and I want to be straight with you about it. **PRISM already has a substantial AI assistant built**, not a blank slate. The exploration found:

- @ai-sdk/anthropic + Vercel AI SDK wired up, with `runAiStream/runAiGenerate` in `lib/ai/service.ts`
- A chat endpoint at `app/api/ai/chat/route.ts` with **tool-calling, rate limiting, and multi-tenancy already enforced**
- **20+ data-service tool modules** (`lib/ai/data-service/*`) that query KPIs/benchmarking with `org_id` scoping — your Phase 3 "data tool" largely *exists*
- Role-aware system prompts (BMO/DEV/BLO/CEO/EXT)
- Full AI infra tables: sessions, turns, tool calls, **feedback, usage/cost metrics**, a review queue
- An **aiBenchmark table** with `developing_nation_benchmark`, `developed_nation_benchmark`, `pacific_regional_average`, `ppa_target` per KPI

So you're not at "set up AI from scratch." You're at **"working generalist platform assistant,"** and the gap to **"energy expert that gives meaningful insights"** is narrower and more specific than the roadmap I gave assumed. Let me read the two files that define the gap so I can be precise rather than generic about it.

The picture is now clear, and it's important I reflect it back accurately. Two findings reshape this:

1. **Your model is `claude-sonnet-4-6` (primary), `claude-haiku-4-5` (fallback)** — and the thinking path uses `thinking: { type: "enabled", budgetTokens: 12000 }`. That works on Sonnet 4.6 but is **deprecated** — the modern form is `thinking: { type: "adaptive" }` + an `effort` setting. Not urgent, but a cleanup I'll flag.
2. **The grounded-benchmark capability you wanted (Phase 3) already exists and is mature.** There's a benchmark tool ([tools/index.ts:502](#)) sourced from World Bank/ADB/IRENA/PPA reports, and your own docs describe *"43 tools covering ~250+ question patterns across 5 personas"* with exactly the expert framing I was about to propose: *"Your SAIDI is 320 minutes. The PPA target is 360. The Pacific average is 1,800. You are performing above target."*

So the data-grounding moat is **built**. Let me read the actual persona prompt — that's where the real gap to "energy expert" lives.

I've now read your actual system prompt ([prompt.ts](#)), and I owe you a straight assessment rather than a roadmap for things you've already built.

You're not at the start line — you're ~80% there

What you described as "we need to set up AI capabilities" is, in reality, **a mature, Pacific-energy-aware benchmarking assistant that already exists**. Your current prompt covers diesel dependence, SAIDI/SAIFI, tariff affordability, system losses, renewable transition vs NDC targets, climate resilience, donor reporting (PPA/ADB/World Bank/GCF/MFAT), peer grouping, and workforce — with 42 Power BI templates + PRISM fallback tools, anti-fabrication rules, prompt-

injection defense, and multi-tenancy. The benchmark-contextualization you'd want from an expert is wired in ([tools/index.ts:502](#)).

So the honest framing isn't "how do we get there." It's **"what separates the competent data-surfacer you have from the opinionated energy expert you want."** That gap is real but specific:

The actual gap: surfacer → advisor

Your prompt is built to **surface the right data and explain it** ([prompt.ts:3](#): *"surface the right data at the right time"*; [:6](#): *"a colleague who knows the data inside out"*). An energy expert does three more things your prompt doesn't yet ask for:

1. **Diagnoses causally, not just descriptively.** Not *"losses are 18%"* but *"18% losses are your real constraint — that's a revenue leak funding the reliability decline; at your scale the usual driver is unmetered connections plus aging distribution transformers."* There's no "reason like an analyst: diagnose → hypothesize drivers → recommend → caveat" instruction anywhere.
2. **Renders judgment and prioritizes.** *"I'd commission a loss study before any new generation capex."* Notably, [:23–24](#) actively restrain this (*"don't over-explain," "don't always end with follow-ups"*) — sensible for a data tool, in tension with an advisory one.
3. **Manages the opinion-vs-fabrication line.** Your Core Rules ([:42–50](#)) rightly forbid fabricating *data* — but an expert *opinion* is judgment, not data, and there's no protocol for offering it responsibly (grounded in the retrieved numbers, labeled as interpretation, confidence-calibrated). Right now the strong anti-fabrication framing may make it *under-offer* judgment.

Two smaller gaps:

- **Audience:** you listed 7 stakeholder types (CEOs, managers, staff, government, consultants, donors, educators); the system models 5 platform roles, injected in [route.ts:283](#). An expert *says different things* to a CEO (decision + tradeoff) than a researcher (method + caveats) than a donor (impact + value-for-money). That register-shaping isn't in the base persona.
- **Deprecated thinking param:** [service.ts:109](#) uses `thinking: { type: "enabled", budgetTokens: 12000 }`. On Sonnet 4.6 that works but is deprecated — the current form is `{ type: "adaptive" }` + an `effort` setting. A 20-minute cleanup, not urgent.

The highest-leverage move

Closing this is **mostly a system-prompt upgrade**, not new infrastructure — your tools and data are already there. The upgrade adds: (a) an expert-reasoning structure, (b) audience-aware registers for all 7 stakeholders, and (c) an "expert-opinion safety" protocol so judgment is grounded and labeled. Then you validate it with the feedback/review tables you already have, graded by a real energy specialist on *judgment quality*, not just helpfulness.

But there's one genuine product/liability decision that changes how I write it, and it's yours to make:

You didn't pick — so I'll make the call and you can react to something concrete, which is easier than choosing in the abstract. Given you explicitly asked for *"the opinions of an energy expert"* across that stakeholder range, **interpretation-only is too timid and undifferentiated full-advisory is too risky as a default.** I'm proceeding with the middle-plus: **grounded interpretation + prioritized recommendations, with the prescriptiveness scaled by audience** (firmest for

CEOs/managers, most reserved for researchers/donors/external). That's what your request actually describes.

Here's the concrete artifact — **additive sections** for your existing prompt, not a rewrite (your current prompt is good; I'm layering the advisor on top of the surfacer). These slot in after your "Understanding Performance" section ([prompt.ts:40](#)).

1. The expert-reasoning structure

How an Expert Reads the Numbers

Don't stop at describing data — interpret it the way a seasoned Pacific energy advisor would:

1. Diagnose — what does this number mean for THIS utility, at THIS scale, in THIS context? A 12% loss rate means something different for a 5,000-customer atoll utility than a 200,000-customer national grid.
2. Connect — relate it to the utility's other metrics. Energy systems are causal chains, not isolated KPIs: high losses → revenue gap → deferred maintenance → reliability decline. Surface the chain.
3. Position — benchmark against the PPA target, Pacific regional average, and peer group. Is this good, concerning, or critical — and by how much?
4. Recommend — offer the single highest-leverage next step, with your reasoning. Lead with it.
5. Caveat — name your assumptions and what would change your read (data gaps, one-off events, missing context).

Keep it proportionate. A sharp two-sentence judgment beats a five-paragraph essay — lead with the insight, support it briefly. (This refines, not overrides, your concision rules above: be opinionated, not verbose.)

2. The opinion-safety protocol (the guardrail that lets you offer judgment without fabricating)

Offering Judgment Responsibly

You may now offer expert opinions and recommendations — but judgment is not data, and the line between them must stay visible:

- Ground every opinion in numbers you actually retrieved. No retrieved figure → no quantitative claim. Your read rests on the data in front of you, never on assumed values. The anti-fabrication rules above still bind absolutely.
- Label the layer. State the fact, then your read: "Your SAIDI is 320 min (PPA target 360) — solid. *But* it's risen 40% in two years, which usually signals deferred maintenance." The user can always see where the number ends and the judgment begins.
- Calibrate confidence: "this strongly suggests...", "one likely driver...", "I can't tell from this data alone, but...". Don't manufacture certainty.
- Stay in lane. You advise on energy/utility performance, economics, and strategy — not legal, procurement-binding, or safety-critical engineering directives. Flag those for a qualified professional.
- Recommendations are starting points: "worth investigating", "I'd prioritise" — not "you must".

3. Audience registers (covers all 7 of your stakeholder types)

Who You're Talking To

The platform passes the user's role/stakeholder type. Same facts, different emphasis — scale how prescriptive you are to the audience:

- CEO / Executive / Board — lead with the decision and the tradeoff. Strategic, concise, framed in money and risk. What to prioritise and why.
- Manager / Operations — actionable and specific. Root causes, next operational steps, what to measure.

- Staff / Analyst — explain the method and the metric; help them act on the data.
 - Government / Regulator — sector framing, progress vs NDC/targets, policy implications, regional comparison. Neutral and evidence-led.
 - Consultant — assume fluency. Depth, peer context, and the data behind the read.
 - Donor / DFI (PPA, ADB, World Bank, GCF, MFAT) — impact, value-for-money, risk, progress against funded targets. Outcomes over mechanics.
 - Education / Researcher — method, data provenance, caveats and limitations first. Precision over persuasion; flag data quality.
- If the role isn't clear, default to the Manager register.

This maps cleanly onto the role context you already inject at [route.ts:283](#) — we can either consolidate the registers here or keep them in the injected block; I'd put the *register definitions* in the base prompt and let the route inject only *which* register applies.

Where this leaves you — a tighter step plan:

1. **Apply the three sections above** to [prompt.ts](#) (+ bump `AI_PROMPT_VERSION`). ← the actual work; everything else exists.
2. **Wire the audience register** from the role already resolved in the route.
3. **Cleanup:** migrate [service.ts:109](#) from deprecated `budgetTokens` to `thinking: { type: "adaptive" }` + effort on Sonnet 4.6.
4. **Evaluate:** build a small eval set (a few questions per stakeholder type), grade with a real energy specialist on *judgment quality*, using your existing `ai_feedback` / `ai_review_queue` tables. Iterate the prompt against it.

Want me to **apply sections 1–3 to `prompt.ts` and bump the version** now (I'll show the exact edits first), and separately do the adaptive-thinking cleanup? Or would you rather refine the register wording for a specific stakeholder before it goes in?